

Institute of Actuaries of India

Subject CP1– Actuarial Practice (Paper B)

November 2020 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable

Assignment 1:**Solution 1:** Reasons for monitoring experience

- Monitoring of experience is fundamental part of Actuarial control cycle
- To check whether actual experience is in line with expectations
- To assess level of profitability/loss arising from such deviation
- To design new control /risk mitigation as well as to
- to monitor the effectiveness of any control or risk mitigation already in place
- To provide information to management
- Use updated information in reserving
- To use updated information for future use like business planning, re-pricing
- To enhance knowledge about customer behavior/market dynamics
- Could be a public disclosure requirement mandated by stock market /insurance regulator
- To understand any trends in experience
- To understand if there has been any change in the risk exposures compared to expected

[0.5 each, **Max. 5 Marks**]**Solution 2:**

Potential reasons of this adverse deviation in loss ratio of XYZ compared to its budget could be

Higher frequency of claims than Budget

- Higher numbers of claims due to un-anticipated disease outbreak in a country or in some geography not anticipated in budget
- Higher than average rate of infection in a year. E.g. Higher frequency of vector borne diseases like Dengue, Malaria in a year than budget

Higher severity of claims than budget

- Due to unanticipated change in interpretation of certain contractual obligations by courts leading to for eg.,
 - Payment of expenses which were thought to be non-payable expenses
 - Coverage of event which was not priced for. E.g. Mental Illness, HIV etc.
- Higher actual medical inflation compared to budget
 - Increase in tariff packages being offered by hospitals was higher than budget
 - Lobbying by hospitals for a higher package across industry which was unanticipated in budget
 - Government increasing minimum tariff on packages covered under government hospitals
 - Increase in drugs prices higher than anticipated

Effect of New business and business mix

- Medium sized company not having geographical diversification could lead to spiral or accumulation of losses in a particular geography
- Budget numbers were allowing for premium/rate hike which did not materialize
- Lower than expected new sales will decrease premium income and increase loss ratio as law of numbers works better on a larger portfolio
- Mix of business could be different from what was budgeted. Higher mix towards higher loss ratio products

- Renewals different than expected- normally in health insurance experience worsens by duration due to expiry of waiting periods initial temporary exclusions etc.,
- Loss of profitable account to competitors
- Introduction of new product which might be mispriced or not allowed for in Budget
- Change in mix of business from distribution channels. E.g. Budget could be having different channel mix than actual business booking for eg., from offline to online etc. impacting claims experience

Operational Reasons:

- Error in calculation of budget numbers (operational risks)
- Budget numbers were on optimistic side
- Increase in frauds /Poor fraud controls by XYZ
- Fraud by agents or distribution channels could lead to high loss ratios. For e.g. there could be a network of agents in collaboration with hospital to gain by making fraudulent claims.
- IT system error which might have caused loss of business or incorrect premium being charged from customer
- Change in Underwriting norms not budgeted for appropriately for eg.,
 - Removal /addition of pre policy health check-up of higher ages
 - Relaxation in underwriting controls to gain business
- Dilution in claims control norms which was not anticipated in Budget
 - Incorrect or higher payment made due to poor claims screening
 - Too much reliance on third party administrator therefore having less control on claims payment
- Undercutting premiums by management to gain or retain market share

Other reasons :

- Different reinsurance arrangements/costs than Planned can impact net loss ratios
- Higher claims provisions than expected. For e.g. Higher reserves toward IBNR due to higher prudence than expected or provisions had been under estimated in budget
- Just a matter of random deviation most likely in case of non-large portfolio
- Regulatory requirement to keep higher provisions than anticipated in Budget
- *Any other valid reason of increase in losses or decrease in premiums*

[12 Marks]

Solution 3:

Impact of adverse deviation on stakeholders of XYZ

- Shareholders
 - Lower profits
 - Might require higher capital to finance
- Board of Directors
 - Concerned about the company's financial soundness
 - May need to relook at Company strategy and business plan
- Government
 - Lower tax due to lower profits
- Regulator
 - Monitoring solvency
 - Ensuring protection of policyholders' interest
- Creditors of company

- Ensuring timely payment of their dues
- Bondholders
 - Ensuring timely payment of coupon or redemption
- Employees
 - Company may cut on jobs, bonuses or pay
 - Lower Provision for pension benefits or retirement benefit
- Policyholders
 - Worried about the protection of their coverage in case of requirement
 - Lower confidence in case of weak financial strength
- Reinsurers
 - May have to pay out higher claims if adverse experience is beyond retention limits
 - May revise reinsurance rates upwards
- Auditors
 - They will have to be extra cautious if financial strength of company is weak
 - They will have to be vigilant about ensuring adequate provisions

[7 Marks]

Solution 4:

Potential reasons of difference in profitability of two insurers offering similar products.

Pricing and underwriting

- Even if products are similar, pricing of two insurance providers could be different.
- ABC might be commanding higher premiums due to brand name
- Geographical mix of customers could be different which could lead to differences in profitability even if prices are similar,
- For eg., Location A could be having higher losses due to higher infection, higher no of hospitals per capita population, higher awareness, rural urban mix etc. The mix difference could lead to different performance
- Initial Underwriting practices could be different
 - Relaxed underwriting in XYZ vs Strict underwriting in ABC
 - Poor controls over pre-policy medical check up –For eg., Medical tests being conducted at poor and unreliable diagnostic centers by XYZ leading to anti-selection and fraud
- Claims underwriting practices/claims management could be different
 - ABC may have better efficient claims management and UW than XYZ
 - XYZ might have relied too much on outsourcing of all claims practices with little intervention or control
 - Same coverage but poor policy wordings by XYZ leading to acceptance of claims that was not was priced for
 - Better fraud control practices in ABC Health Insurer. Better tools for screening of proposals /claims and efficient usage of technology
 - Poor claim defense practices by legal team of XYZ leading to higher losses on legally contest cases

Target market and distribution

- Target mix of customer could be different.
 - ABC might be targeting and attracting better risk profile, better socio economic profile,
 - Average size of policies could be higher for ABC leading to improved expense recovery and less cross subsidy

- ABC might be sourcing business from different distribution channels where the performance is favorable. For e.g. Higher presence in digital space by ABC vs only traditional channels by XYZ which enables ABC to attract a different socio-economic profile
- ABC might have collaboration with strong distribution partner. For e.g. ABC might be selling their policy to a customer of strong banking partner and is able to procure higher volumes of profitable business (for eg., large ticket size policies).

Lower expenses due to

- ABC has negotiated better package rates of hospitalization from network hospitals
- Higher expenses by XYZ on managing business
 - Higher salaries or large workforce
 - Higher cost of IT systems
 - Offices in high rent areas
 - Poor IT infrastructure, requiring significant maintenance cost YoY.
- Tax efficiency by ABC in carrying forward losses and better managing tax outgo
- Higher acquisition cost of securing business
 - High commission payouts
 - High bonuses or inefficient sales reward system
- Investment performance & Expertise
 - ABC might able to earn higher investment income than XYZ due to superior investment policy and performance
 - XYZ might have suffered losses on poor investments decisions
 - Write off of corporate bonds to which XYZ have exposure
- *Any other valid reason*

[14 Marks]

Solution 5: Advantages of having a standardized product

- Company has pricing flexibility and can charge as per their experience, exposures and risk appetite
- Since every insurer will have to offer a same product, it will provide more information regarding pricing or risk selection practices of competitors.
- It will add to customer base of insurer and gives more opportunities to cross sell other products
- Companies get an opportunity to distinguish themselves through better service offerings rather than competing on obscure policy terms and conditions
- It will increase the overall premium size of health industry and will benefit insurers
- Companies can now negotiate better tariff rate with hospitals since large business for insurer will mean higher volumes
- Having a standardized product will force companies to strengthen risk selection/underwriting practices. This will enhance over all capabilities which will indirectly have impact on rest of the business also
- Coverages of standardized products might not be same as existing products and thus it might not have much impact on existing business. For e.g. Company might be offering a feature rich health plan and having a standardized product might provide new opportunity with different targeted customer base.
- Higher business will lead to better utilization of existing resources and will lead to expenses optimization
- Might not require significant increase in sales force/costs to sell a standardized health product

- Company can design their strategy to gain market share in rural areas by this product. Since the government is promoting it will have a higher trust factor and easier to sell business in areas where there is little presence of insurer
- Companies can offer their enriched product in urban markets by promoting standardized cover as base vanilla product with minimum coverages.
- To offer upgrade to customer of standardized product at renewals
- Decline in rejection ratio of claims due to better awareness of coverages of standardized products
- Image building by promoting government standardized product

Disadvantages of having a standardized product

- It might lead to lower innovation across the industry
- It will eat up share of other sophisticated products of insurer
- Companies will invest less in training resources, agents, claim team etc. if this product becomes the sales leader which will inhibit any improvements as well
- Value addition brought by distributor would be less and hence his compensation could decrease
- The reverse could also be true that now distributor has to be compensated more for selling the more sophisticated products
- IT system needs to be configured to manage new product along with existing products
- Since every company will offer same product, premium rate will end up showing the inherent inefficiencies, poor risk management practices of insurer;
- Could lead to companies compromising on their internal risk management processes to reduce costs and hence premium rate
- Building any differentiation other than price for eg., in terms of service will take money
- Risk of high adverse selection
 - Existing product might be having more rating factors than standardized product
 - This could lead to selection by customers where company is charging higher premiums but now cannot charge in standardized product
 - This could have an adverse effect on whole health portfolio
 - Company might have to align rating factors of other health products in line with standardized products
- Difficulties in administration
 - Claims team will have to process claims with different policy conditions
 - Underwriting criteria may need to be customized for this product
- Companies will not be able to charge higher premiums in standardized product than existing superior coverages products even though risk of new product is unknown
- Company may have to withdraw their existing products if their sales fall
- It will lead to higher loss ratios or lower profitability due to adverse selection or risks of mis-pricing or lower premium realization per life
- Existing products of company with higher proportion of customers without any restriction on coverage will need rate correction sooner than anticipated if new business sales fall for existing products. This will further accentuate the problem due to gap in premium of two products.
- Existing customer might want to shift to cheaper standardized product at renewals
- Reduce return of capital of shareholders

[0.5 Mark each, unless stated otherwise, **Max. 12 Marks**]

[50 Marks]

Assignment 2**Solution 6:**

- As a result of the regulation the insurer will be forced to break the fixed deposits with the bank though it can be done over a 3 year period.
- There is likely to be penalty on breaking the fixed deposits which will be a direct hit to the insurer
- It may be able to negotiate a lower penalty with bank if the quantum of investment is large
- The bank may also be happy to reduce the penalty if the lending rates have considerably fallen since the time investment was made.
- This should be the case as interests have fallen as mentioned.
- Since the insurer has to break the deposit it will have to make investment in other asset classes though it can be done gradually over a 3 year period
- There could be some transactional cost associated with investing in other asset classes
- Tax implications needs to be considered
- Changes would be required in the investment policy of the company in light of regulation change
- This will be required to be approved by the Board
- It has to invest the quantum received by breaking the fixed deposit as per the revised investment policy of the company
- The supervisory reserve are likely to get impacted as the yield in the asset backing the liabilities is likely to change
- Any fall in the interest rate used to discount the liabilities will lead to increase in the mathematical reserves
- This will lead to lower profit for the insurer in the year when the change was made
- The increase in the reserves is likely to lead to fall in the solvency margin of the company
 - On account of fall in available solvency margin due to lower profits due to penalties for early closure of deposits, if any, higher transaction costs etc.
 - Higher required solvency margin due to increase in reserves
- At an extreme level, it could also make insurer insolvent
- leading to need for additional capital support from shareholder/other sources
- This could also lead to reputation risk for the insurer
- Thereby impacting the new business
- and could also lead to poor persistency
- It could invest in risky but high yielding assets
 - This could lead to lower financial impact if
 - As insurer is allowed to take credit for higher yields while deriving valuation interest rate

[0.5 each, **Max. 12 Marks**]**Solution 7:**

- Restrict exposure to each individual bond
- The duration of such bonds could be low compared to liability duration, which could lead to an ALM mismatch and a necessity to hold a mismatch reserve

- Use a wide spread of different corporate bonds, for diversification
- Allow for appropriate level of corporate default rate while arriving valuation interest rate
- Monitor the credit risk exposure regularly
- have regular discussion in the management meetings to sensitise management on risks
- Validate the ratings provided by rating agencies
- This could be done internally on ongoing basis by employing credit analysts
- Do proper due diligence regarding the financials of the company issuing the bonds
- In particular before buying large quantities of corporate bonds from particular entity to examine any stress
- Use credit derivatives to protect against defaults
- The credit derivatives can itself lead to credit risk
- Risk can be mitigated using the collateral requirements
- The company should ensure that it maintains adequate liquid assets to allow for unexpected liquidity requirements
- E.g.:- cash or short term debt or high liquid bonds
- This needs to consider the volatility of non-maturity claim payment timings like surrenders
- The company should aim to match its cash in-flows and out-flows as closely as possible
- Liquidity risk should be included in the company's risk policy
- and requirements for monitoring, measuring, reporting
- and limiting the liquidity risk should be set out
- The company should use scenario analysis to assess when there could be a liquidity issue
- It could put an emergency short term cash access arrangement in place

[0.5 each, **Max. 11 Marks**]

Solution 8: Mortality heavier

- Assumption was inadequate or
- Target market different from expected
- Mortality improvement lower than expected
- Poor underwriting or Risk selection
- Claims management poorer than expected
- However the underwriting and claims management are likely to have lesser impact in case of single premium plan
- CAT event - Geographical concentration of risks
- The data may not be credible and therefore investigation may not be fully relied on
- Particularly if there are outliers in the actual claims
- Low allowance for IBNR is made in the claim experience used in pricing which led to underestimation of mortality rate
- Few large claims are distorting the picture if mortality experience is being done by Sum Insured/Sum at risk
- Anti-selection, though likely to be limited in SP products, but could be being done to avail tax benefits (for eg. Avoid estate tax)

[Max 6]

Surrender lighter

- The assumptions at pricing might have been incorrect
 - Taken with reference to different customer profile

- Unrelated product
- target market differences
- Guarantees are valuable
- Particularly since interest rates have fallen
- And the investors buying the guaranteed products are likely to be risk averse
- The economy is doing well leading to lower need for surrendering the policies in general
- Tax issues of surrendering the policies
- Surrender penalties are high
- Retention practices employed by the company like
 - Regularly informing clients about the benefits of continuing the policies
 - Advertising it's financial strength
 - Conducting interactions with customers coming for surrender and offering alternatives like a policy loan
 - Predictive modelling and pre-emptive action etc

[0.5 each, Max. 6]

[12 Marks]**Solution 9:**

Cyber risk commonly refers to any risk of financial loss, disruption or damage to the reputation of an organization resulting from the failure of its information technology systems. [2]

Perils covered-

- Business interruption loss could be covered
This would cover any loss of income if business has been interrupted due to a cyber-attack
- Coverage for any investigation costs into attacks
- Insurer could indemnify the costs of repair to a damaged website
- Extortion may also be covered
- i.e.- pay a ransom to a hacker threatening to destroy website/release sensitive data
- Coverage for the costs of notifying customers of a security or privacy breach
- There will also be costs relating to any claims of infringement of privacy following a breach
- Cost of rebuilding computer system/website following damage say from virus attack
- Cost of restoring company data
- Liability risks- Any damage to third party due to onward transmission of a virus
- Pay for remedial PR costs
- Pay for legal costs
- Fine imposed by regulator
- Loss of intellectual property

[0.5 each ,Max. 8 Marks]

Solution 10:**Controls include:**

- Ensuring anti-virus software is in place throughout the organisation and maintained up to date
- installing firewalls
- implementing appropriate user access rights to hardware and software systems
- restricting employee internet access to those with a business need
- adopting policies and procedures to reduce exposure to harmful viruses
- not permitting employees to use personal internet based email accounts
- restricting software that can be installed on company computers
- carrying out appropriate training for employees so they are aware of the policies and procedures
- measures to prevent data being stolen- putting in place safeguards to prevent downloads of mass data to removable storage devices
- Working with appropriate third parties to store/protect very sensitive data
- Working with third parties/IT system auditors to test the effectiveness of security procedures
- Regular IT audits
- Testing versatility of BCP programs in case of business interruption due to cyber attack
- Regular scheduled data backups
- Secure mobile devices- laptops, black berry, mobile phones through encryption,
- Regular updates and patches to application /programs
- Allow use of only licensed software in all devices

[0.5 each ,**Max. 7 Marks**]
[50 Marks]
